

Chapter 1

Historical developments of NMR logging

1.1 Introduction

The application of nuclear magnetic resonance (NMR) to petrophysics for studying the properties of water and/or oil saturated rocks dates back to the early 1950's. By the 1960's, nuclear magnetic resonance logging tools using the earth's magnetic field were built for oil exploration. However, these tools did not gain wide popularity due to various limitations. It was not until the early 1990's that pulsed NMR type logging tools¹ were introduced and gained a wide acceptance by the petroleum industry. Since then, numerous applications have evolved along with significant tool improvements. Users can now extract information, not only on the saturating fluids in the rocks, but also on the pore size distributions, thus the pore fabric, of different rock types.

The information we obtain from the NMR properties of the fluid-saturated rocks is primarily based on the nuclear spin relaxation. NMR spectroscopy used in the identification of chemical states is seldom used in the studies of rocks. While there are numerous books that deal with NMR spectroscopy, very few provide a thorough account of NMR relaxation in fluid-saturated porous media and its applications to petroleum exploration. It is the primary objective of this book to fill that gap.

¹Logging tools are instruments lowered into a drilled borehole in the earth formation to make physical measurements for hydrocarbon or mineral prospecting.

1.2 Nuclear magnetic resonance

The phenomenon of nuclear magnetic resonance was first observed independently in 1946 by scientists at Stanford (Bloch [22, 23]) and Harvard (Purcell et al. [181], Bloembergen et al. [24]) Universities. The subject quickly grew into a sophisticated technique with wide applications in physics and chemistry. Many of these applications have to do with investigations of microscopic interactions in materials at the molecular and atomic levels. Intricate interactions often manifest themselves in shifts of nuclear magnetic resonance frequency. Thus, NMR becomes a powerful tool for the identification of chemical states in matter, a technique commonly known as NMR spectroscopy.

NMR also found applications in biomedicine, food, and petroleum industries. In the biomedical field, where NMR is used as an imaging tool for medical diagnosis, there has been extensive development in the last decade. The application of NMR for petroleum exploration started as early as 1950's. Scientists at Mobil, Exxon, Shell, Texaco, and Chevron were among the early pioneers in the use of NMR technology [226]. Many of the early applications of NMR spectra were investigations of the composition of petroleum and petroleum fractions. Chevron, for some time, maintained an extensive program on the application of NMR to petrophysical studies of rocks and fluids and in collaboration with Byron-Jackson and Borg-Warner, and in consultation with Varian Associates, developed a well logging tool for formation evaluation².

The NMR signal from hydrogen nuclei is proportional to the population of hydrogen atoms in the sample, and the signal relaxation rate is proportional to the viscosity of the fluid. Therefore, it was natural to consider using these NMR properties for oil exploration. However, early investigators found that the nuclear magnetic relaxation of water in rocks is much faster than that in the bulk phase. The increase of the relaxation rate is primarily due to the surface relaxivity and is related to the surface to volume ratio of the pore space. This discovery led to the demonstration of the connection between the NMR properties of porous media and their permeability. The early research work in Chevron [194, 212, 37, 216] as well as others (e.g. Schlumberger, Mobil, Texaco, Borg-Warner, and Varian [123]) not only resulted in a number

²Formation evaluation is a branch of earth sciences for evaluating petrophysical properties of the earth formation from measurements of many logging tools.

of publications illustrating the important applications of NMR technology for oil exploration, but also led to the development of an experimental logging tool [38] and to commercial logging services in the 1960's by Pan Geo Atlas Corp. (PGAC) [66] and by Schlumberger [100]. These efforts finally led to the extensive developments of the modern day NMR logging technology (Miller et al. [163], Prammer et al. [176], Kleinberg et al. [124], Freedman et al. [87]).

The commercial Nuclear Magnetism Logging (NML) tools, when and where available, operated by the free precession in the earth's magnetic field until 1994, when the first modern NMR logging tools operating at higher magnetic fields became commercially available after several years of field testing. The modern day logging tools carry their own magnets, and operate at higher Larmor frequencies. Each service company has its own commercial name (Halliburton/Numar's is called Magnetic Resonance Imaging Log (MRIL), Schlumberger's is called Combinable Magnetic Resonance tool (CMR)).

1.3 NMR logging tool developments

The initial idea for a NMR logging tool came from Russell Varian [36], and subsequent feasibility studies were conducted at Varian Associates from 1953 to 1955. At that time, a number of oil companies as well as Schlumberger started to do important work on NMR in fluid-saturated porous media. The initial thought was that one could use the Nuclear Magnetism Logging tool to measure hydrogen in both oil and water in porous rocks, and they would be distinguishable by different relaxation times, with water usually having the longer relaxation time. Since water and oil have about the same hydrogen density, one would expect to obtain a measurement of the total porosity of the rock as well.

Subsequent developments were largely unexpected. Nobody anticipated that the surface effects on relaxation of water in the pore space would be so significant. Water in the pores of most rocks was found to have a wide range of relaxation times due to surface relaxivity. The oil relaxation time in the rocks, on the other hand, was found *usually* to have the same relaxation time as that in the bulk phase³. Because the oil relaxation time is likely to be the

³Note, for oil wet rocks, oil also behaves similarly to water, having a wide range of relaxation times. The fact that oil relaxation time in rocks is usually the same as that of the bulk phase is because most rocks are water wet. (See discussion in Chapters 3 and 4)

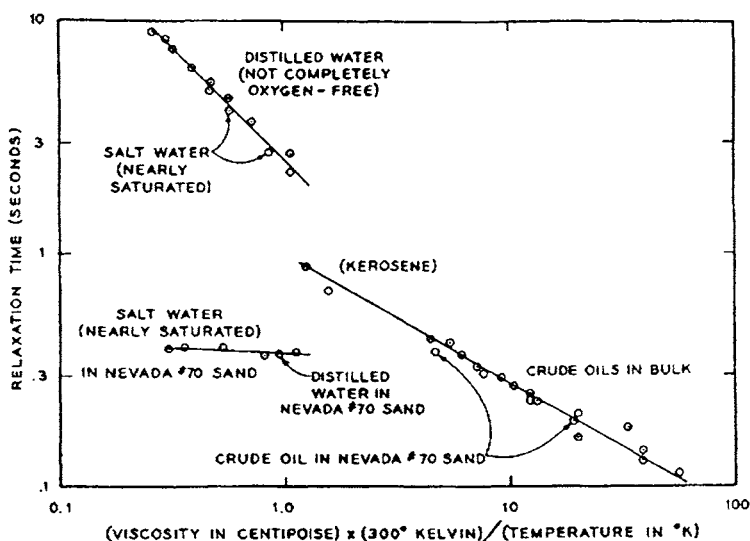


Figure 1.1: Relaxation times in bulk fluids as well as those in the porous media. (From Brown [36].)

same as for some of the water in the rock, it becomes a nontrivial matter to differentiate oil from water simply using their different relaxation times.

Figure 1.1 shows the understanding of relaxation times of various fluids in bulk phase and in porous media at the end of 1955 [36]. Relaxation times were plotted against viscosity divided by absolute temperature. The bulk water and various bulk oils show longer relaxation times for lower viscosity. An important observation is that putting oil in the pores of a sand has very little effect on its relaxation time, whereas putting water in the pores of a sand greatly reduces relaxation times and almost eliminates the dependency of relaxation time on viscosity or temperature. This behavior is found to be due to the surface relaxation effect in the pore space, with water being in contact with the pore surface, whereas the oils usually not.

In addition to the complexity of not being able to differentiate water and oil in the rocks, the early development of NMR ran into another obstacle. The tool was designed based on the observation of proton free precession in the earth's magnetic field at a frequency of about 2 kHz. The coil (shown schematically in Figure 1.2) was 1.75 m long and 15 cm wide. In operation, the coil was energized with a strong dc current to produce a field for a few

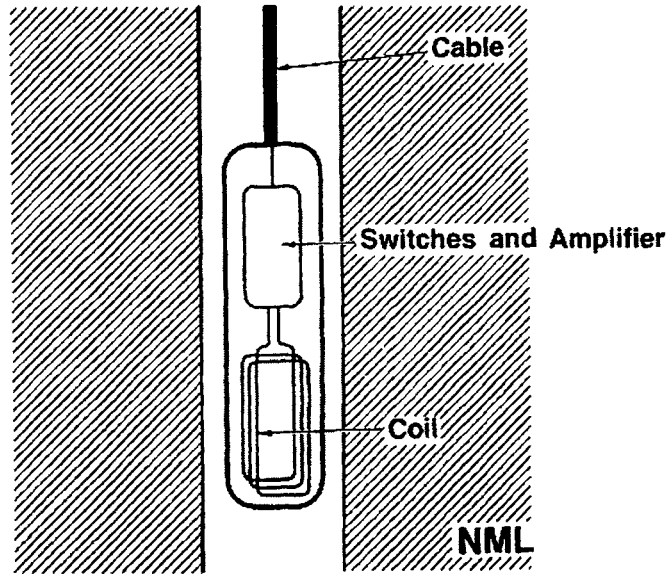


Figure 1.2: Early Chevron design of a nuclear magnetic logging tool. (From Brown [36].)

seconds while a nuclear magnetization was built up. Then the current was abruptly turned off, and the same coil was used to receive the signal from the proton free precession in the earth's magnetic field. The instrument dead-time was about 30 ms (later reduced to 21 ms). As a result, part of the water in the pore space with relaxation times less than the dead time was not measurable. Thus the NML tool would not be able to distinguish the water and oil in pore space; it would not even be able to measure the total porosity.

With these new understandings, the development of the NML was continued with different objectives. Petrophysical studies in the laboratory have shown that the relaxation times of water are closely related to surface to volume ratio of the pore space. Large surface to volume ratio has small permeabilities, and vice versa. Thus, the NMR measurements can be used to predict permeability. Fortuitously, the dead time was quite close to the T_2 cutoff established for irreducible water saturation. The fluid that was measurable by NML was called the "Free Fluid". This term continues to be used now to describe the portion of the fluid in the pore space that is movable, as "Free Fluid Index" (or FFI).

An NMR tool was finally developed based on the earth's field concept (Brown and Gamson [38]) and was used in field measurements [100]. But the tool did not gain wide acceptance due to several limitations. A major problem was that the tool measures the signals from the water and oil in the formation, which are of interest to us, as well as the signals from the water-filled borehole, which are not. Thus, to kill the signals from the borehole, powdered magnetite (magnetic iron oxide) was added to the borehole fluid. This made locally inhomogeneous fields that shortened signal lifetimes so much that signals were not measurable with a 30 ms dead time instrument. However, this procedure is frequently undesirable due to other operational constraints, and it also excludes applications in deep wells. The sonde housing must be non-magnetic, which limits its use in hostile environment. Besides, the tool sensitivity depends on the tool orientation with respect to the earth's magnetic field. This creates complications in the signal calibration.

To circumvent the problems associated with using the earth's magnetic field, people started thinking of using permanent magnets to generate a homogeneous field at an external location outside the probe, the so-called remote inside-out concept. Cook [65] and Cooper and Jackson [67] independently suggested a method for producing a toroidal region of homogeneous magnetic field outside the NMR probe. In the design by Jackson and his colleagues [47, 112], two cylindrical magnets are placed in a tool housing with their north poles opposing each other. These magnets generate a static field which is radially symmetric. There exists a toroidal region on the horizontal mid plane perpendicular to the central axis where the radial component of the magnetic field H_r can be made homogeneous to any chosen degree over prescribed regions.

Figure 1.3(a) shows the magnet configuration for the production of a toroidal region of homogeneous radial magnetic field. Figure 1.3(b) shows the three dimensional plot of the radial component of the magnetic field as a function of the radial distance and the separation between the opposing magnets. Notice that for a fixed separation, i.e., h/a (where h is the distance between the opposing magnets, a is the radius of the cylindrical magnets, and r is the radial distance from the origin at the mid plane between the magnets), the radial component of the magnetic field, H_r , goes through a maximum. In the neighborhood of this maximum, the field can be made uniform over a certain region, to any desired accuracy, by increasing the separation of the opposing magnets, h/a . The inset of Figure 1.3(a) indicates a coil, when

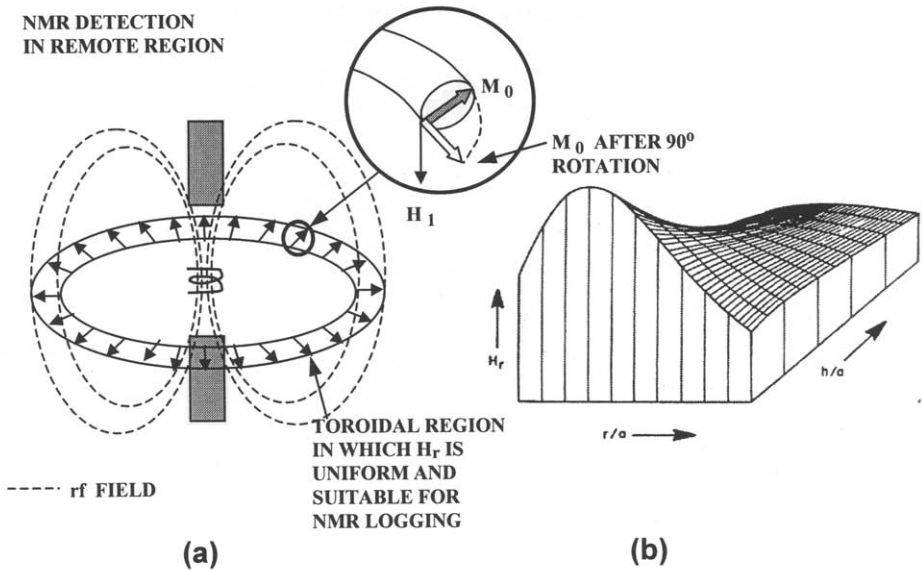


Figure 1.3: (a) Toroidal region of homogeneous field of two opposing cylindrical magnets (From Jackson et al. [112]). (b) Three dimensional plot of the radial field H_r versus magnet separation h/a and radial distance r/a . (From Cooper and Jackson [67].)

placed in the middle of the opposing magnets, that can generate a 90° pulse in the sensitive region. Following the application of the radio frequency (rf) pulse, the magnetization precesses in a plane perpendicular to the magnetic field H_r . The coupling of the precessing magnetization and the coil allows the NMR signal to be detected. For a detailed understanding of how the tool works, the readers are referred to Chapter 2.

Later on, Clow et al. [62] built a NMR logging tool based on Jackson's and his colleagues' design. They placed in the copper solenoid a core of a high permeability ferrimagnetic material, such as manganese-zinc ferrite, thus greatly increasing the Q factor and improving the signal to noise ratio.

A basic difficulty with the Jackson design is that the location and field strength, of the toroidal region of highly homogeneous field, tend to vary over time during operation, due to changes in local earth's magnetic field, formation temperature, and mechanical parameters of the field producing permanent magnets. When the rf coil is tuned to a fixed frequency, it lacks the flexibility to resonantly match the varying field condition.

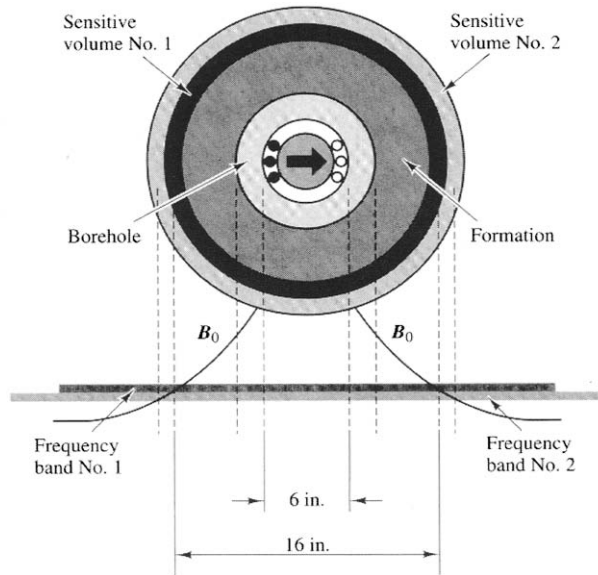


Figure 1.4: The cross sectional view of Numar/Halliburton's Magnetic Resonance Imaging Log (MRIL) tool based on Shtrikman-Taicher's design. The tool is centered in the borehole, and the probed region is a thin cylindrical shell. (Courtesy of Numar/Halliburton Corp.)

Shtrikman and Taicher [200] came up with a different design in which a long cylindrical magnet is magnetized transversely to its long axis, producing a static magnetic dipole field in the earth formation which varies as $1/r^2$. The rf coil is wound lengthwise on the cylindrical magnet in the plane containing the magnetization direction and the longitudinal axis (Figure 1.4). The resonant condition is satisfied on a thin cylindrical shell coaxial with the magnet. The probed region of this thin cylindrical shell has a length roughly equal to the long dimension of the coil along the axis.

In the Shtrikman-Taicher design, the magnetic field tapers off as $1/r^2$. The tuned rf coil will always select a thin cylindrical shell whose field strength matches the resonance frequency, thus alleviating the resonance matching problem faced by the Jackson design. The Shtrikman-Taicher design was later developed into the Numar/Halliburton's Magnetic Resonance Imaging Log (MRIL) tool. One of the weaknesses of this design is the smallness of the probed dimension in the radial direction, approximately 1 to 2 mm thick, and the field is not homogeneous, having a well-defined field gradient. Such a

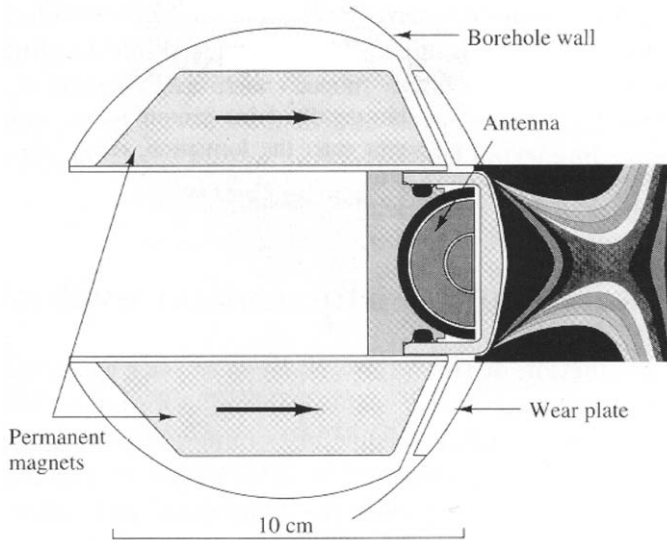


Figure 1.5: The cross sectional view of Schlumberger's Combinable Magnetic Resonance (CMR) tool. The sonde is pressed against the borehole wall. The probe region has a saddle-point configuration. Its center contains an area with relatively homogeneous magnetic field. (Courtesy of Schlumberger Ltd.)

weakness is compensated by the long dimension of the cylindrical shell, which results in a large probed volume, and a consequent adequate signal to noise ratio. The presence of a field gradient, initially thought to be a drawback, turned out to be a desirable feature for the application of hydrocarbon typing (See discussion in NMR log applications in Chapter 5).

In contrast to the centralized design of Shtrikman and Taicher, scientists in Schlumberger (Kleinberg et al. [124]) designed a pad-type tool. A cross-sectional view of their design is shown in Figure 1.5. The sonde consists of three slab magnets magnetized in the same direction, pressed against the borehole wall, resulting a homogeneous field in a region about 0.75 in inside the formation with a width about an inch in the radial direction and about 6 in along the the borehole wall. The antenna is filled with ferrite to increase its sensitivity. Such a design has a high depth resolution and a low sensitivity to the borehole fluid salinity. It was later developed into Schlumberger's Combinable Magnetic Resonance tool (CMR).

The NMR logging tools currently on the market, Halliburton/Numar's

MRIL and Schlumberger's CMR, have gone through considerable improvements over their original designs. The Logging While Drilling (LWD) tool for NMR measurements is also being introduced (Prammer et al. [179]). In the foreseeable future, improved and new tools will be introduced, making an even greater impact on the formation evaluation in the estimation of the quantity and producibility of hydrocarbon reserves.

1.4 Applications in formation evaluation

Since the introduction of the pulsed NMR type logging tools, investigators have found numerous applications of NMR logs in formation evaluation. The most frequently used applications include: mineralogy-independent porosity estimates, irreducible water saturation determination, permeability prediction, hydrocarbon typing, oil viscosity estimation, and other applications, such as residual oil estimation, thief zone identification, mineralogy typing, etc.

The mineralogy-independent porosity estimate is an important application of NMR logs. The traditional method of porosity estimates uses either gamma rays (density logs) or neutrons (neutron logs). Both methods depend intrinsically on the mineralogy of the formation. They frequently require mineralogical correction to arrive at an accurate porosity determination. In a complex mineralogical environment, such corrections are not straightforward, and NMR logs can be used to find the solution. However, NMR porosity is not always mineralogy independent. Where the formation contains a significant amount of ferromagnetic and/or paramagnetic materials, the NMR logs could underestimate porosity because some of the T_2 relaxations are too rapid to measure.

The irreducible water saturation determination, permeability prediction, hydrocarbon typing, and oil viscosity estimation, are unique applications offered by NMR logs. No other logs provide similar kinds of information. The initial estimation of these quantities using NMR logs are quite rudimentary. As our understanding increases, the tool capability improves, and the methods of analysis become more sophisticated, the estimation of these quantities becomes more and more accurate.

We shall explain each of these applications in later chapters. The book is organized in the following manner. We begin with a brief introduction of the basic principles of nuclear magnetic resonance. Once we understand

the basic meaning of T_1 and T_2 , and how we measure them, we proceed to describe their relaxation behaviors for various fluids of interest to the petroleum industry. This is followed with a discussion of NMR properties of fluid-saturated rocks as measured in the laboratory, in preparation for the understanding and interpretation of NMR logs. Then, various NMR log applications are discussed with examples.

A chapter devoted to NMR data inversion is presented. The nature of the ill-posed inversion problem is discussed briefly. The book ends with a rigorous theoretical treatment of NMR relaxation behavior and diffusion of spins in fluid-saturated porous media. In the applications of NMR logs, various approximations must be made to achieve quantitative estimates of the desired quantities. This theory is presented to show some of the limitations of these approximations.

To simplify the description of this very complicated subject, we shall use classical physics to illustrate the NMR phenomenon. The quantization of spin angular momentum and the energy transition in relaxation processes are, however, quantum phenomena. There are a few places where one has to resort to quantum mechanical explanations which are not intuitive. For a more advanced treatment of NMR, the reader is referred to Abragam [1] and Slichter [201].